

Ben Boland

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Professional Summary

Embedded systems and real-time software engineer with hands-on experience in firmware development, hardware/software co-design, and real-world robotics systems. Background includes Rust, C, Java, microcontrollers, GPU programming, and custom hardware projects. Interested in embedded and performance-critical systems where software directly interacts with hardware.

Education

Illinois State University

Dec 2025

B.S. in Cybersecurity

- **Coursework:** Linux & Windows Administration, Network Infrastructure, Cryptography, Penetration Testing

Projects

Semi-Autonomous Golf Course Divot Filler

Apr 2025 – Sep 2025

Personal Project

- Designed and implemented a **real-time perception-driven robotics system** with position and orientation tracking using wheel odometry.
- Developed **embedded Rust firmware** on an **STM32F407**, controlling motors and servos with deterministic timing guarantees.
- Integrated **DMA-driven SPI communication** to reduce CPU overhead and improve real-time responsiveness.
- Built a low-latency 2D visualization and instrumentation tool to debug perception, control loops, and system geometry in real time.
- Refactored firmware architecture to use **FreeRTOS**, separating logging, control, and hardware I/O into independent tasks, improving scheduling determinism and system maintainability.
- Owned the full system lifecycle: **hardware design, firmware, real-time software, mechanical design, and performance optimization**.

16-bit 5-Stage RISC CPU

Sep 2023 – Mar 2024

Personal Project

- Designed and built a 16-bit 5-stage pipelined RISC CPU using discrete logic, implementing instruction decode, hazard handling, and pipeline control.
- Developed a complete **assembler and tooling in Rust** to support instruction generation and software execution.

Java Game Modding: Modified a game's rendering pipeline to support runtime-swappable shaders, enabling highly configurable visual effects.

CUDA Password Cracker: Implemented a GPU-accelerated password cracking tool using CUDA to parallelize hash computations across thousands of threads.

Web Application (Academic): Led backend development of a cloud-deployed system; experience with APIs, authentication, and CI/CD

Multiplayer Voxel Engine (In Progress): Architecting a performance-focused engine in Rust, emphasizing memory efficiency, profiling, and throughput under constrained resources.

Skills

Languages: Rust, C, TypeScript, Java, Python

Embedded & Hardware: STM32, DMA/SPI/I2C/UART, microcontroller firmware, custom PCBs

Systems & Debugging: FreeRTOS, multithreading & concurrency, low-latency pipelines, profiling, Linux CLI

Tools: Git, Docker, Wireshark, Tracy Profiler